



A handwritten signature in black ink, featuring a large, stylized capital letter 'R' with a loop, followed by a horizontal line extending to the right.

A handwritten signature in black ink, featuring a large, stylized initial 'R' or 'B' followed by a horizontal line extending to the right.

$$N = 100 \pm$$

$$N = 100 \pm$$

$EC(v)$

v

a, b, c, d, e, f, g

SoC_i i

E_{bat}

d_i i

v_i i

t

t

t

M

CV

σ

μ

N

$\bar{h}_r$ r

Δh_r r

$h_{,r}$ r

$t_{i,r}$ ir

N_r r

B_r r

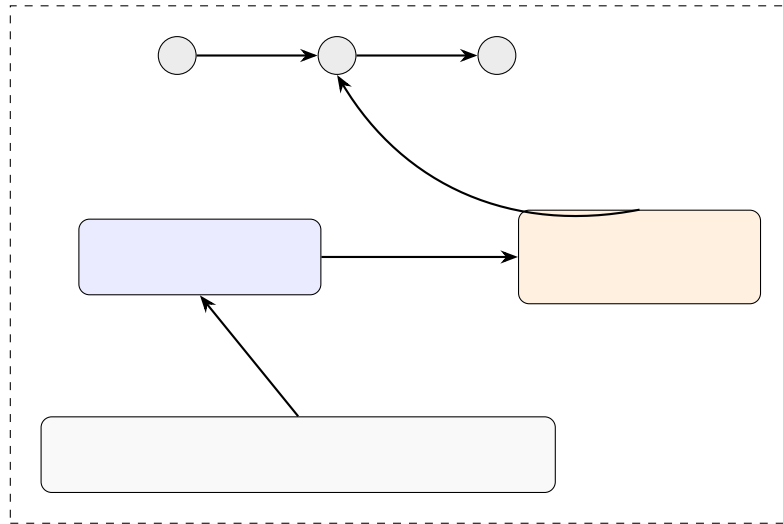
SoC_b b

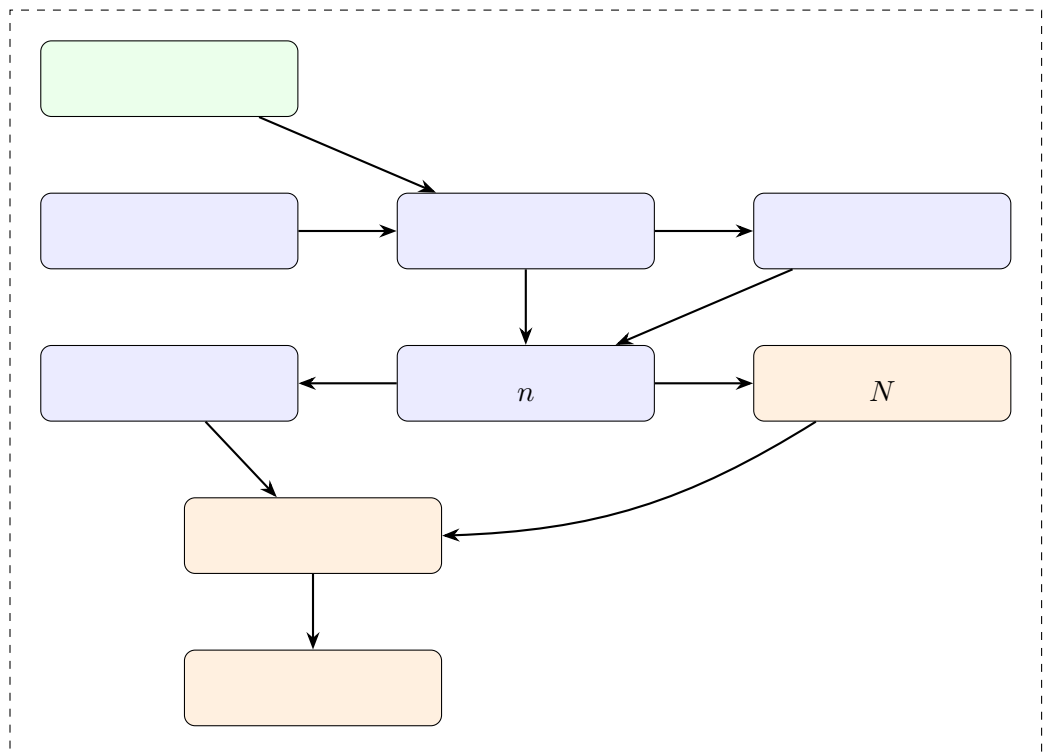
$w_{q,b}$ b

t b

t

Q





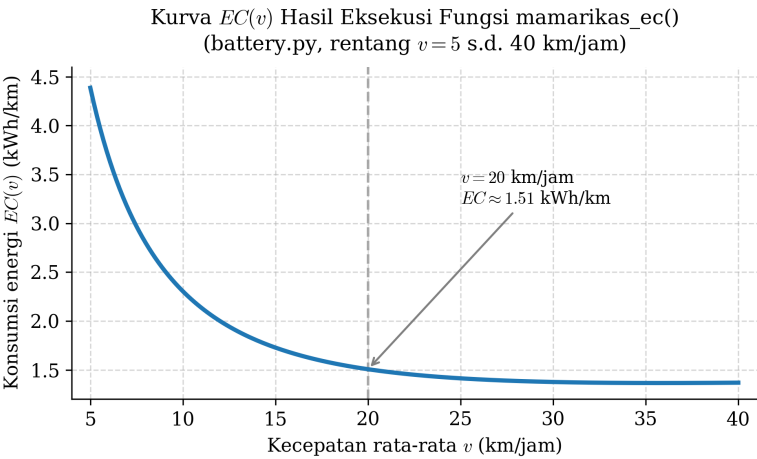
ECv

$$EC(v) = \frac{av^2 + bv + c + \frac{d}{v}}{ev^2 + fv + g}$$

$$a = 0,0185b = 1,40 \times 10^{-8}c = 6,91d = 1,54e = 9,64 \times 10^{-6}f = 2,79 \times 10^{-4}$$

$$g = 1,15 \times 10^{-4}$$

$EC(v)$ 540matplotlibliblab/visualizations.ipynbmamarikas_ec()battery.py
 d/v



$EC(v)$ 540

$EC(v)$

$$SoC_i = SoC_{i-1} - \frac{EC(v_i) \times d_i}{E_{bat}} \times 100$$

$$d_iiv_iE_{bat}SoC \leq 30SoC20$$

$$v^2d/vvE[EC(v)] > EC(E[v])$$

$$\begin{aligned} CV &= 15 \\ CV &= 5 \end{aligned}$$

tt

$$\begin{array}{l} t=t\times M\\ \sigma=\sqrt{(1+CV^2)}\\ \mu=(t)-\frac{\sigma^2}{2}\\ t\sim(\mu,\sigma)\\ \left(\frac{d}{40}\right)\leq t\leq\left(\frac{d}{5}\right) \end{array}$$

$$d/v$$

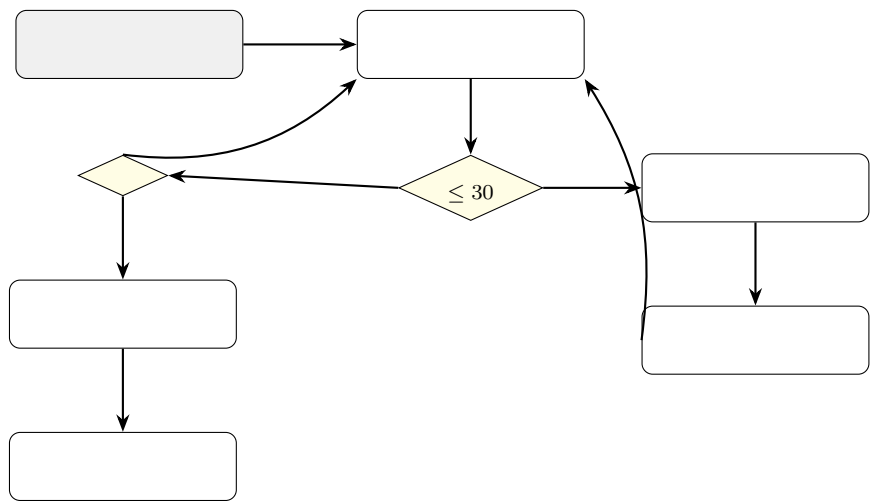
$$\text{simulation_result.json}$$

$$\begin{array}{l} \bar{h}_r\Delta h_r\\ \bar{h}_r=\frac{1}{N_r-1}\sum_{i=2}^{N_r}(t_{i,r}-t_{i-1,r})\\ \Delta h_r=\bar{h}_r-h_{,r} \end{array}$$

$$\begin{array}{l} t_{i,r}irN_r\\ (r)=\left\{\begin{array}{l} ,\quad \exists b\in B_r: SoC_b\leq 20\vee\\ (w_{q,b}\geq 30\wedge t\geq t) \end{array}\right., \end{array}$$

$$rbw_{q,b}\geq 30tt$$

$$SoC\leq 30$$



$$\geq 15$$

$$T_{datang} D_{normal} JS$$

$$T_{selesai} \leftarrow T_{datang} + D_{normal}$$

$$D_{overlap} \leftarrow [T_{datang}, T_{selesai}] JS$$

$$D_{overlap} \geq 15$$

$$NN = 50N = 100N = 50N = 50N = 100$$

$$Q_{spklu} B_{transit}$$

$$Q_{proyeksi} \leftarrow Q_{spklu}$$

$$b \in B_{transit}$$

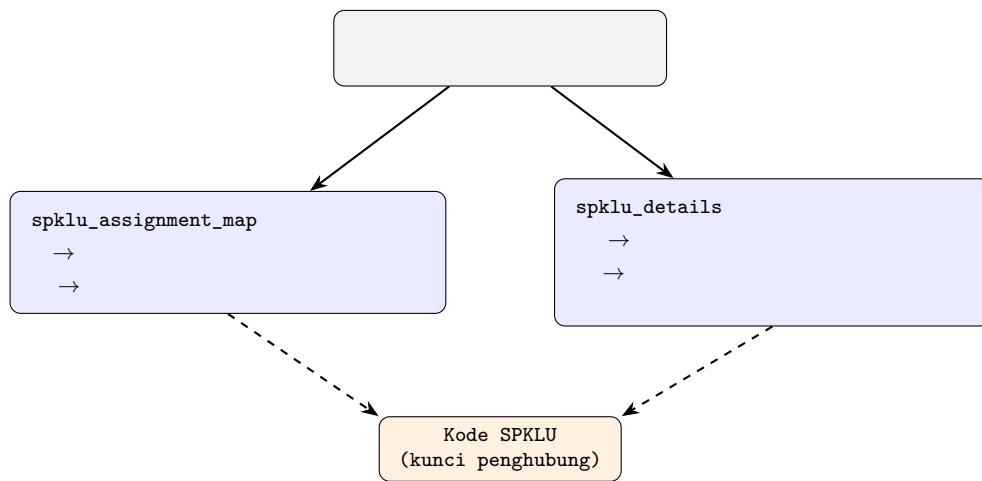
$$b \approx$$

$$Q_{proyeksi} \leftarrow Q_{proyeksi} + 1$$

$$Q_{proyeksi} \geq$$

```
{
  "scenario_id": "s17b_probabilistic_4loc_prop",
  "charging_mechanism": "full_opportunity",
  "extra_bus_per_route": 0,
  "dynamic_charging": false,
  "spklu_assignment_map": {
    "SPKLU-ADI": ["1A", "14"],
    "SPKLU-NGB": ["1B", "3B", "6", "9", "10", "11", "13",
"15"]
  },
  "spklu_details": {
    "SPKLU-ADI": { "n_gun": 2, "travel_min": 60, "
charge_min": 60 },
    "SPKLU-NGB": { "n_gun": 8, "travel_min": 30, "
charge_min": 60 }
  }
}
```

spklu_assignment_mapspklu_details



assignment_map details

heapq

simulation/battery.pyBatteryDrainModelmamarikas_ec()

```
def mamarikas_ec(self, v: float) -> float:
    """
    Hitung konsumsi energi (Wh/km) berbasis kecepatan rata-rata (v)
    menggunakan Persamaan 1 dari Mamarikas dkk. (2025).
     $EC(v) = (a \cdot v^2 + b \cdot v + c + d/v) / (e \cdot v^2 + f \cdot v + g)$ 
    """
    v = max(v, 5.0) # Pengaman matematis (v_min=5 km/jam)

    a = 0.0185
    b = 1.40e-8
    c = 6.91
    d = 1.54
    e = 9.64e-6
    f_coef = 2.79e-4
```

```

g = 1.15e-4

numerator = a*(v**2) + b*v + c + d/v
denominator = e*(v**2) + f_coef*v + g
return numerator / denominator

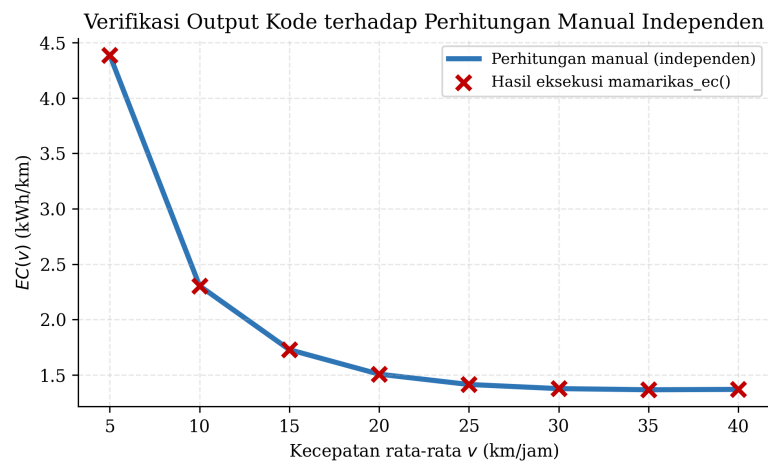
```

battery.py

```
v = max(v, 5.0)d/vmamarikas_ec()
```

DrainResultsoc_tracebreach_hard_limit

```
mamarikas_ec()5400
```



mamarikas_ec()

```

daily_time_modifierrandom.uniform(0.8, 1.2)CV = 15engine.segment_cv
= 0.05mamarikas_ec()

```

simulation/engine.pyheapq

BatteryDrainModelSPKLUResourceHeadwayEvaluator

charging_mechanism

```

spklu_resourceBusAgent._run_trip()spklu_resourcedynamic_charging=True
dynamic_version="v2"

```

$$\begin{aligned}
&B \\
&Q \\
&b \in B \\
&\quad Q.(\langle t, , b \rangle) \\
&Q \\
&\quad \langle t, , b \rangle \leftarrow Q.() \\
&\quad == \\
&\quad \quad b \\
&\quad \leq \\
&\quad \quad Q.(\langle t, , b \rangle) \\
&\quad \leq 20 \\
&\quad \quad b \\
&\quad \quad Q.(\langle t, , b \rangle) \\
&\quad == \\
&\quad \quad b \\
&\quad \quad tw_q \\
&\quad \quad w_q \geq 30t \geq t \\
&\quad \quad Q.(\langle t + , , b \rangle) \\
&\quad == \\
&\quad \quad Q.(\langle t + , , b \rangle)
\end{aligned}$$

SPKLUResource

scenario_id

N

$NN = 50N = 100$

$$NS$$

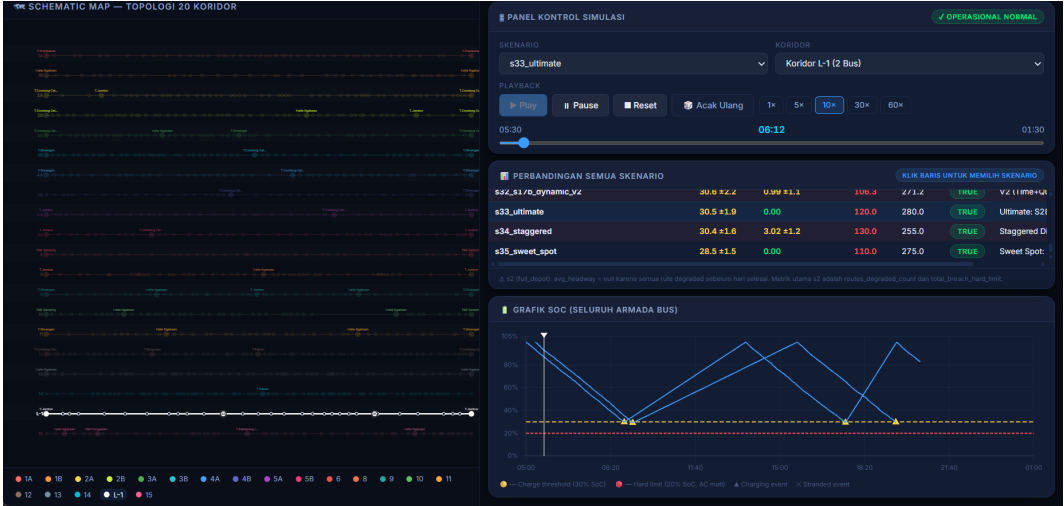
$$R \leftarrow \emptyset$$
$$i = 1N$$
$$M$$

$$i$$

$$R$$

$$\mu \sigma R$$





$N = 100$

$\pm \text{headway_dishub_min}$

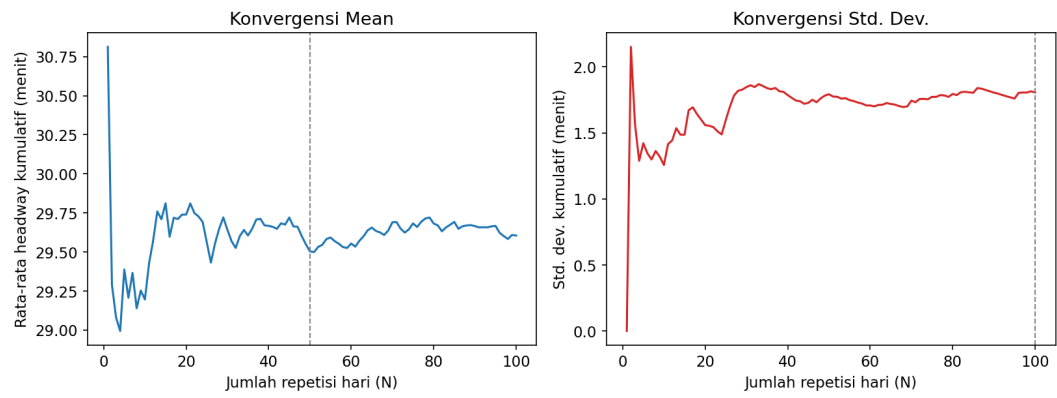
headway_reliable

$N = 100$

N

N		

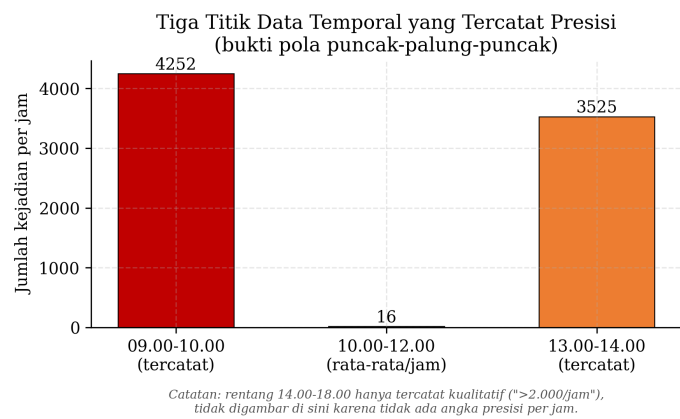
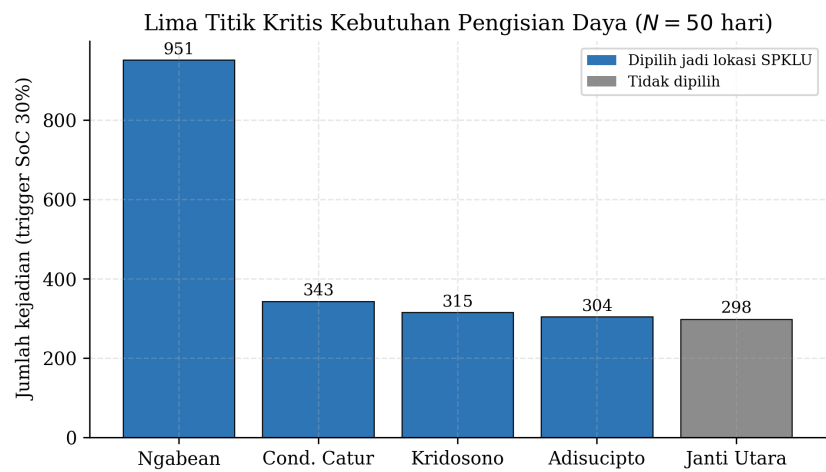
$N = 50N = 100N = 50N = 75N = 100N = 50N = 100$



N

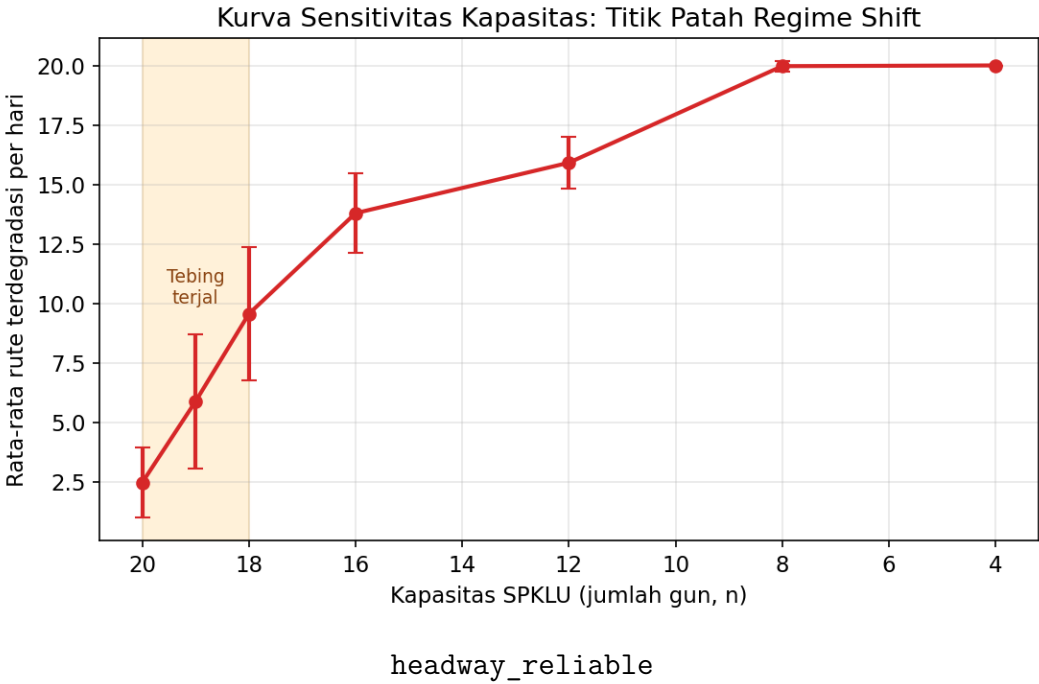
$N = 50$

		$32,16 \pm 2,21$	2,47
		$35,55 \pm 2,06$	6,97
		$40,84 \pm 2,19$	11,44
		$34,93 \pm 1,81$	3,03



headway_reliable

		$32,16 \pm 2,21$	2,47
		$33,22 \pm 2,93$	5,52
		$36,71 \pm 2,78$	8,85
		$44,78 \pm 4,68$	12,73
		$70,88 \pm 9,12$	15,45
		$38,69 \pm 5,28$	19,90
			20,00



$r = -0,845$

$\leq 0\pm10$

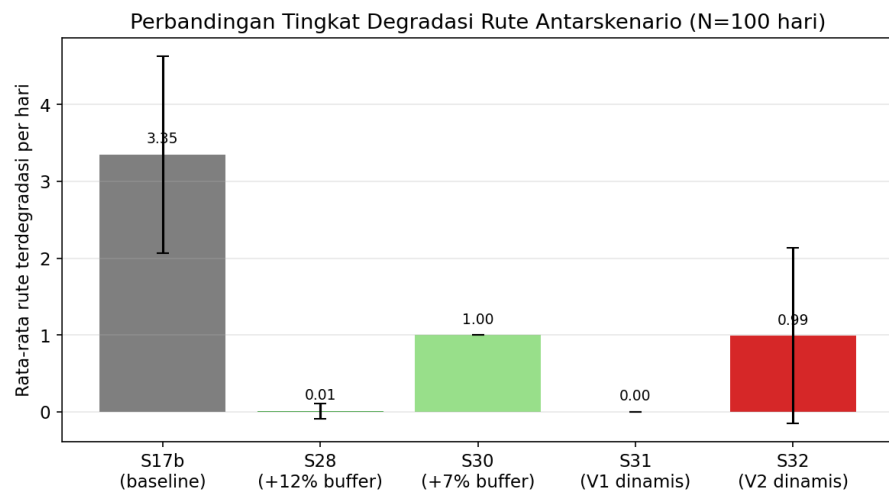
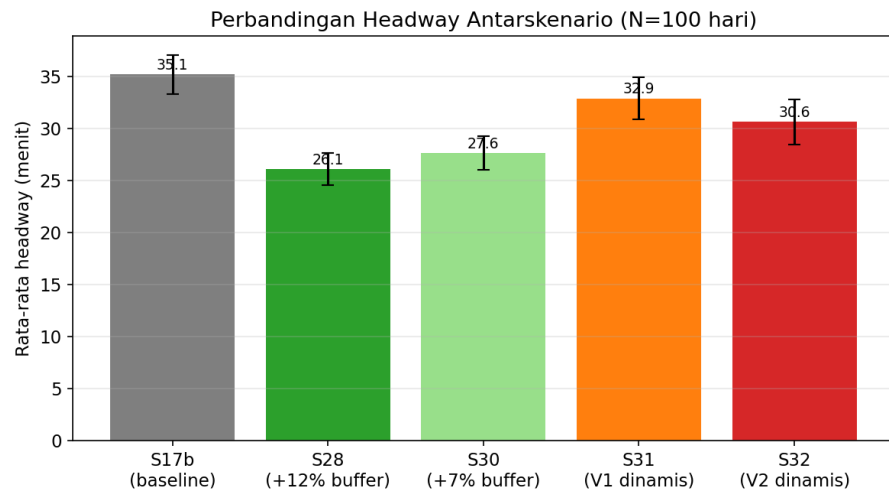
$$N = 100$$

	$34,93 \pm 1,81$	3,03		
	$27,62 \pm 1,57$	1,01		
	, $\pm$,	0,01		
	$33,10 \pm 1,91$	,		
	$30,52 \pm 2,12$	0,74		

$$N = 100$$

	$30,78 \pm 1,98$	0,00		
	$30,65 \pm 1,53$	3,00		
	, $\pm$,	,		

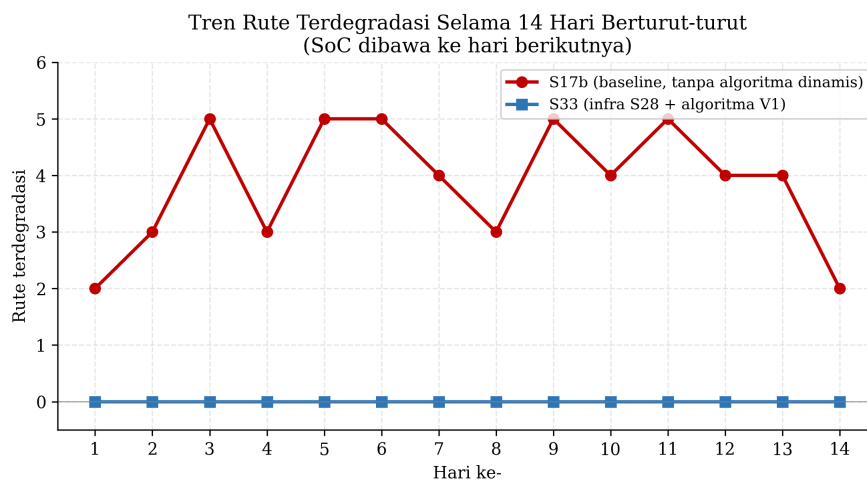
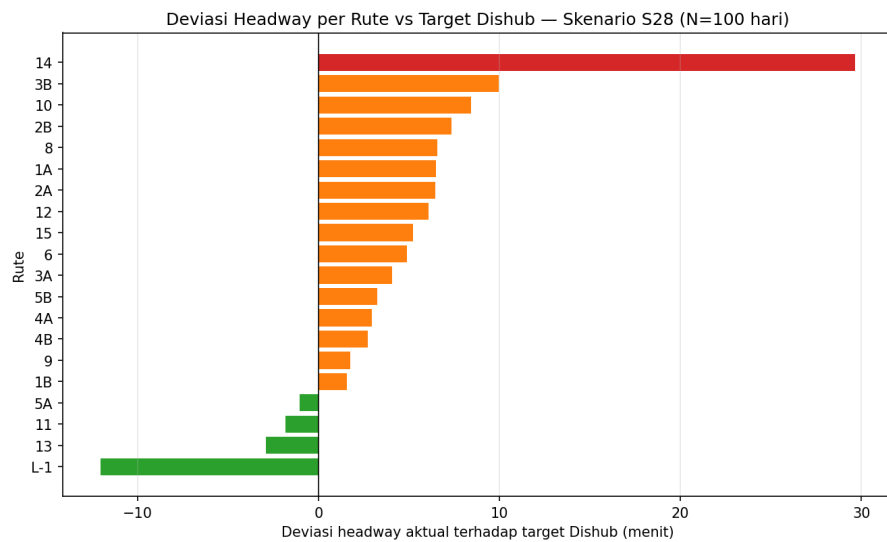
$$\pm 10$$



$34,93 \pm 1,813,03$

			-1,35	
			-1,84	
			-2,86	
			-12,21	

$$n_{gun} = 19n_{gun} = 18$$



$N = 100$

$\pm \text{headway_dishub_min}$

[illegible]

$N = 100$	$26,14 \pm 1,52$	$30,78 \pm 1,98$	
	0,01	0,00	
			$\pm$
		0,00	

±


```

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$$EC(v) = (a \cdot v^2 + b \cdot v + c + d/v) / (e \cdot v^2 + f \cdot v + g)$$

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    v = max(v, 5.0) # Pengaman matematis (v_min=5 km/jam)

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    f_coef = 2.79e-4
    g = 1.15e-4

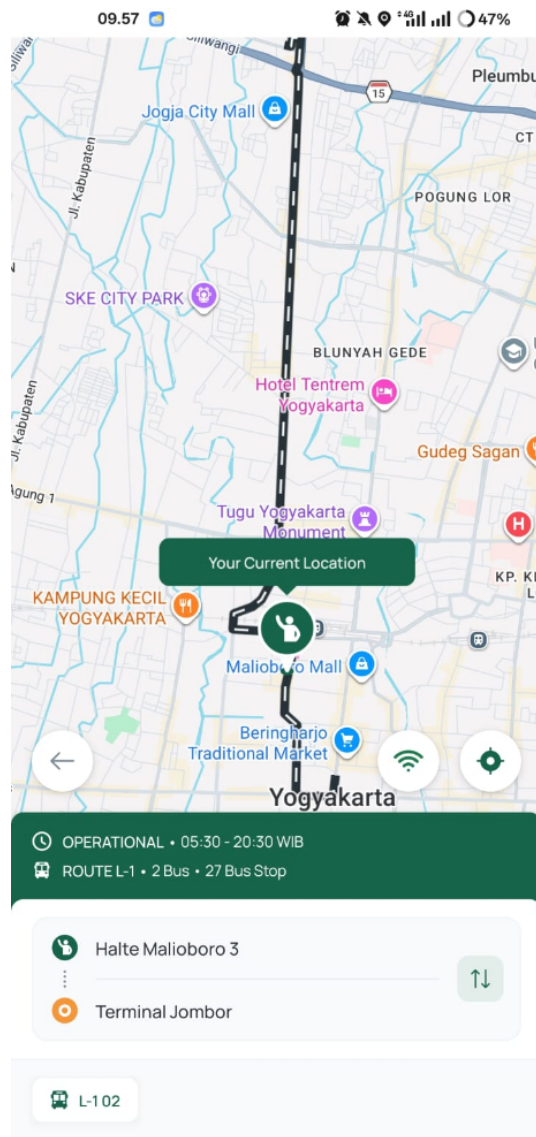
    numerator = a*(v**2) + b*v + c + d/v
    denominator = e*(v**2) + f_coef*v + g
    return numerator / denominator

```

v

engine.py









$$N = 100 \pm N$$

[illegible]

